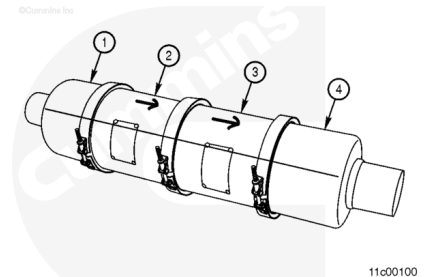


Trainings ▾

General Information

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst element can result.



Due to the number of exhaust aftertreatment applications, this procedure has been written to be generic. Illustrations within this procedure will **not** represent all applications. The aftertreatment system is composed of four sections. These sections are:

1. Inlet
2. Catalyst
3. Aftertreatment diesel particulate filter
4. Outlet.

Note : In some applications, the catalyst can be integrated into the inlet of the exhaust aftertreatment.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable first.



- See equipment manufacturer service information for aftertreatment system removal instructions. The original equipment manufacturer (OEM) will designate whether

aftertreatment system components are to be removed as individual components or as a complete system for service.

- Mark the direction of exhaust flow on the outside of the aftertreatment system to aid in installation.
- Mark the orientation of the exhaust gas particulate trap differential pressure sensor to aid in installation.
- Disconnect the vehicle batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

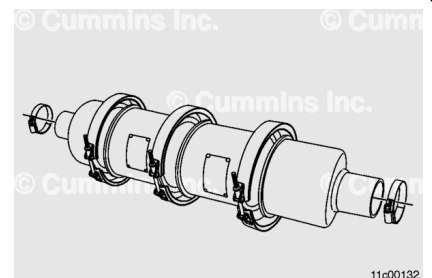
Remove

System - Remove the straps or bolts securing the aftertreatment system to the vehicle chassis. Refer to the OEM service manual.

Remove the v-band from the inlet, and the Torca™ clamp from the outlet of the aftertreatment system. Refer to Procedure 011-048 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Disconnect the aftertreatment diesel particulate filter differential pressure sensor wiring harness connector. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-443 in Section 19.
(/qs3/pubsys2/xml/en/procedures/16/16-019-443.html)



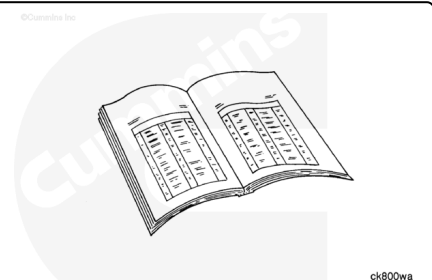
Component - To remove the aftertreatment inlet or outlet sections: Refer to Procedure 011-048 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

To remove the aftertreatment diesel particulate filter and the joined catalyst - aftertreatment diesel particulate filter: Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Remove the catalyst: Refer to Procedure 011-049 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)



Disassemble

Use this procedure to remove the aftertreatment inlet or outlet sections. Refer to Procedure 011-048 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Use this procedure to remove the catalyst. Refer to Procedure 011-049 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)

Use this procedure to remove the aftertreatment diesel particulate filter. Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Use this procedure for aftertreatment diesel particulate filter cleaning instructions. Refer to Procedure 011-041 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

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Clean and Inspect for Reuse

Inspect inlet or outlet canister. Refer to Procedure 011-048 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Inspect the aftertreatment diesel particulate filter canister. Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Inspect the catalyst canister. Refer to Procedure 011-049 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)

Inspect the exhaust gas temperature sensors. Refer to Procedure 019-449 in Section 19

(/qs3/pubsys2/xml/en/procedures/16/16-019-449.html)

This procedure is found in the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control System, ISX and ISM Engines, Bulletin 4021560.

Inspect the differential pressure sensor. Refer to Procedure 019-443 in Section 19.

(/qs3/pubsys2/xml/en/procedures/16/16-019-443.html)

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This procedure is found in the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control System, ISX and ISM Engines, Bulletin 4021560.

Inspect the differential pressure sensor tubes. Refer to Procedure 011-047 in Section 19.

(/qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html)

Inspect the aftertreatment system wire harness.

Inspect the Torca™ clamps and/or V-Band clamps for leaks. Refer to Procedure 010-024 in Section 10.

(/qs3/pubsys2/xml/en/procedures/10/10-010-024-tr.html)

This procedure is found in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239.

Refer to Procedure 010-024 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-024-tr.html)

This procedure is found in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322.

Assemble

Install the aftertreatment inlet or outlet sections. Refer to Procedure 011-048 in Section 11.

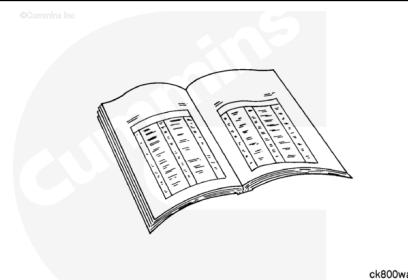
(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Install the aftertreatment diesel particulate filter. Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Install the catalyst. Refer to Procedure 011-049 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)



Install

Note : Make sure that the aftertreatment system is oriented so that the aftertreatment diesel particulate filter differential pressure sensor is installed in the same orientation noted during disassembly.

Note : The aftertreatment system must be installed so that the aftertreatment diesel particulate filter differential pressure sensor tubes slope downward to drain condensation away from the differential pressure sensor.

Note : Make sure that the differential pressure sensor tubes are not making contact with each other or any other vehicle components prior to tightening the Torca™ clamps or V-Band clamps .

System - Install the straps or bolts that secure the aftertreatment system to the vehicle chassis. Refer to the OEM service manual.

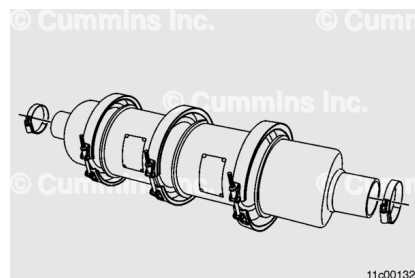
Install the inlet and outlet of the aftertreatment on to the exhaust pipe. Refer to Procedure 011-048 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Note : If a Torca™ clamp is replaced during service, be sure to replace it with another Torca™ clamp. Do not use a u-bolt as a replacement clamp. U-bolt clamps can crush the tailpipe and make it difficult to remove the aftertreatment system for future service.

Tighten the Torca™ clamps used to secure the inlet and outlet of the aftertreatment to the exhaust pipe. Refer to Procedure 011-048 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

Connect the exhaust temperature sensor to the wiring harness connectors. Refer to Procedure 011-048 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

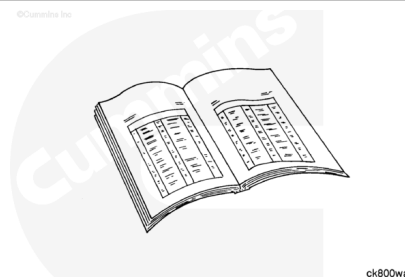
Connect the aftertreatment diesel particulate filter differential pressure sensor wiring harness connector. Use the following procedure from the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-443 in Section 19.
(/qs3/pubsys2/xml/en/procedures/16/16-019-443.html)



Component - To install the aftertreatment inlet or outlet sections: Refer to Procedure 011-048 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

To install the catalyst: Refer to Procedure 011-049 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)

To install the aftertreatment diesel particulate filter or joined catalyst-aftertreatment diesel particulate filter: Refer to Procedure 011-041 in Section 11.



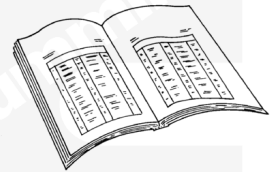
Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable first.

- Connect the batteries. Refer to the OEM service manual.
- If the aftertreatment system was being serviced for an aftertreatment diesel particulate filter cleaning, use INSITE™ electronic service tool to clear related fault codes.

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